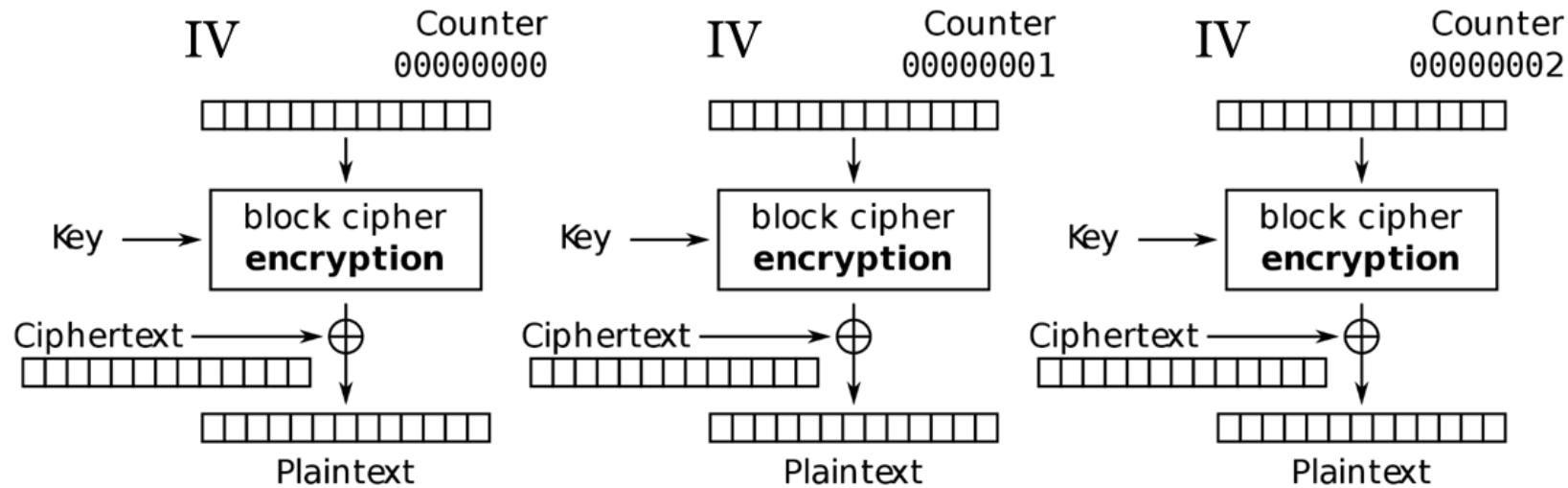


Message authentication code (MAC)

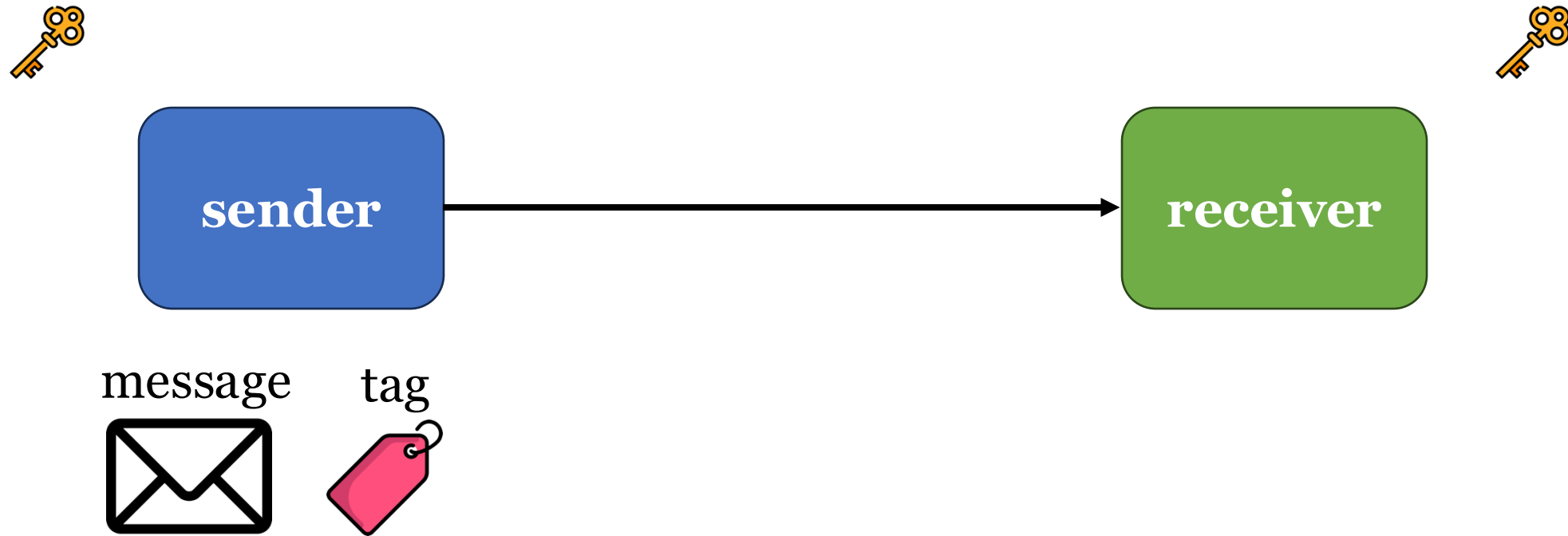
Secrecy vs. Integrity

- Secrecy does not imply integrity
- Integrity is equally important



Counter (CTR) mode decryption

Message authentication code (MAC)



Verify ✓ or ✗

Note message is in plaintext, we only care about integrity now!!!

Message authentication code (MAC)

- $k \leftarrow \text{Gen}(1^n)$
- $t \leftarrow \text{MAC}_k(m)$ (τ is also commonly used for tag)
- $\{0,1\} \leftarrow \text{Verify}_k(m, t)$

Correctness: for every $k \leftarrow \text{Gen}(1^n)$, every message m ,

$$\text{Verify}_k(m, \text{MAC}_k(m)) = 1$$

MAC security attempt 1

Experiment $\text{MAC}_{A,\Pi}(n)$:

1. A key k is generated using $\text{Gen}(1^n)$
2. A outputs (m, t)
3. Experiment outputs 1 if $\text{verify}_k(m, t) = 1$

MAC security attempt 2

Experiment $\text{MAC}_{A,\Pi}(n)$:

1. A key k is generated using $\text{Gen}(1^n)$
2. Adversary A is given input 1^n , and **oracle access to $\text{MAC}_k(\cdot)$** .
3. A outputs (m, t)
4. Experiment outputs 1 if $\text{verify}_k(m, t) = 1$

MAC security (unforgeability)

Experiment $\text{MAC}_{A,\Pi}(n)$:

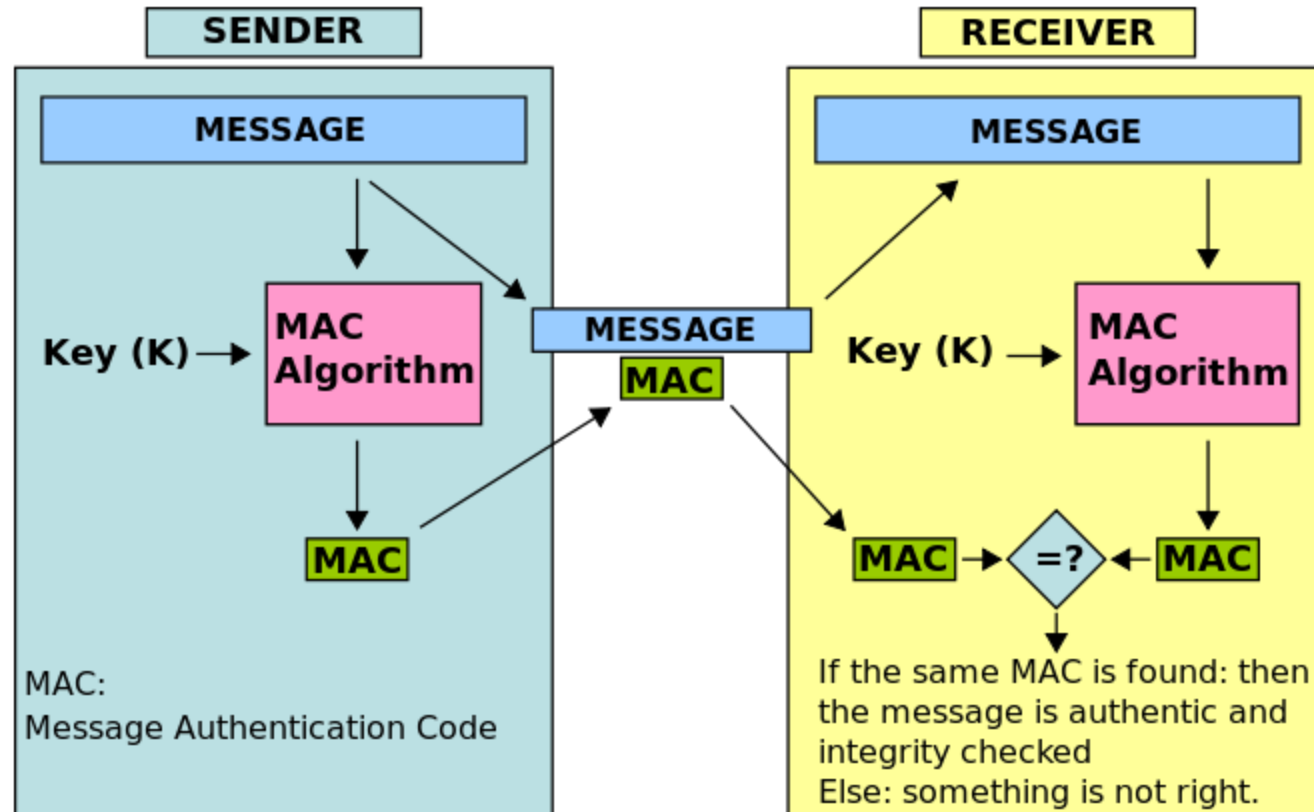
1. A key k is generated using $\text{Gen}(1^n)$
2. Adversary A is given input 1^n , and **oracle access to $\text{MAC}_k(\cdot)$** . Let Q be the set of all queries made by A
3. A outputs (m, t)
4. Experiment outputs 1 if (1) $\text{verify}_k(m, t) = 1$ and (2) $m \notin Q$

Definition: a Mac $\Pi = (\text{Gen}, \text{MAC}, \text{Verify})$ is **unforgeable/secure**, if for all probabilistic polynomial time adversaries A , there is a negligible function negl such that for all n ,

$$\Pr[\text{MAC}_{A,\Pi}(n) = 1] \leq \text{negl}(n)$$

Strong unforgeability

- A can't generate a different MAC for a queried message
- Deterministic MAC and canonical verification



Construction 1: fixed-length MAC

Let F be a PRF

- $k \leftarrow \text{Gen}(1^n)$
- $\text{MAC}_k(m): t = F_k(m)$
- $\text{Verify}_k(m, t):$ compute $F_k(m)$, check if $t = F_k(m)$

Security

- Theorem: If F is a PRF, then Construction 1 is a secure MAC
- Proof: build a D to distinguish $F_k(x)$ from $f(x)$
 1. Run $A(1^n)$. Whenever A queries $\text{MAC}_k(\cdot)$, compute $t = f(m)$ and send it to A
 2. Get (m^*, t^*) at the end from A .
 - Compute $f(m^*)$
 - If $t^* = f(m^*)$ and A never queried $\text{MAC}_k(\cdot)$ on m , output 1. Otherwise, output 0.

$$\Pr[D^{F_k(\cdot)}(1^n) = 1] = \Pr[\text{MAC}_{A,\Pi}(1^n) = 1]$$

$$\Pr[D^{f(\cdot)}(1^n) = 1] = 2^{-n}$$

MAC security (unforgeability)

Experiment $\text{MAC}_{A,\Pi}(n)$:

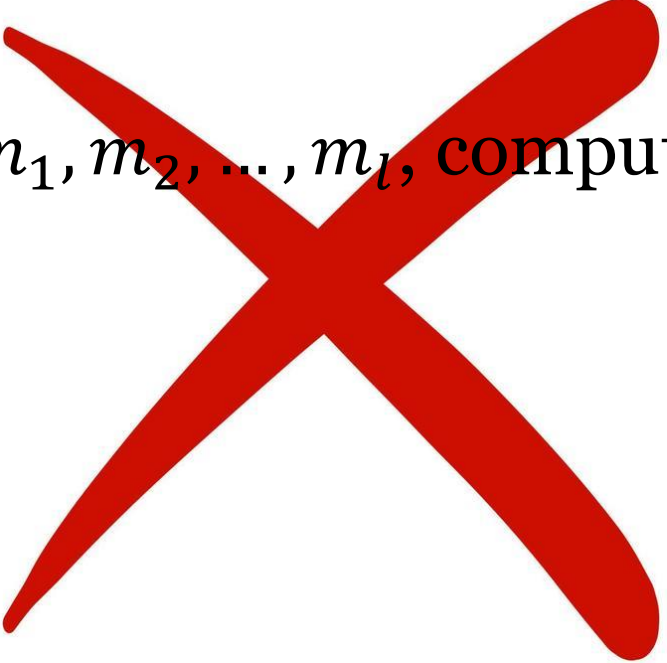
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Definition: a Mac $\Pi = (\text{Gen}, \text{MAC}, \text{Verify})$ is **unforgeable/secure**, if for all probabilistic polynomial time adversaries A , there is a negligible function negl such that for all n ,

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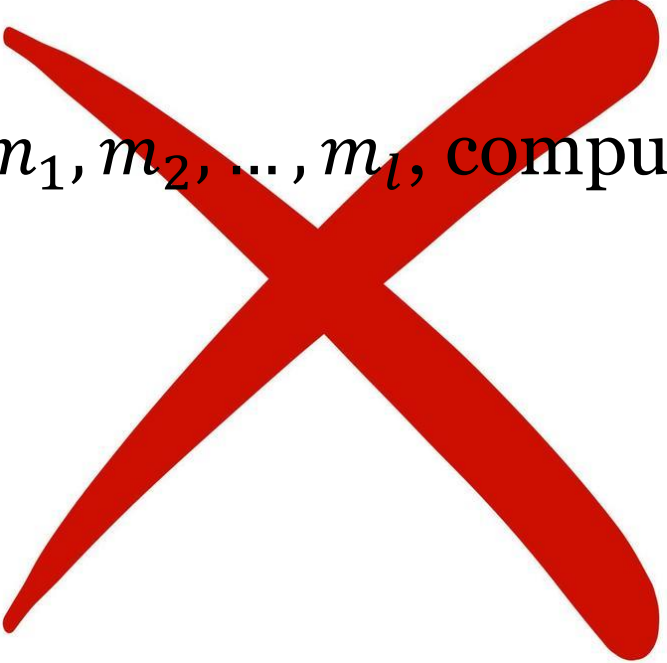
Longer messages?

Idea 1: parse m as $m_1, m_2, \dots, m_l$, compute $t_i = F_k(m_i)$



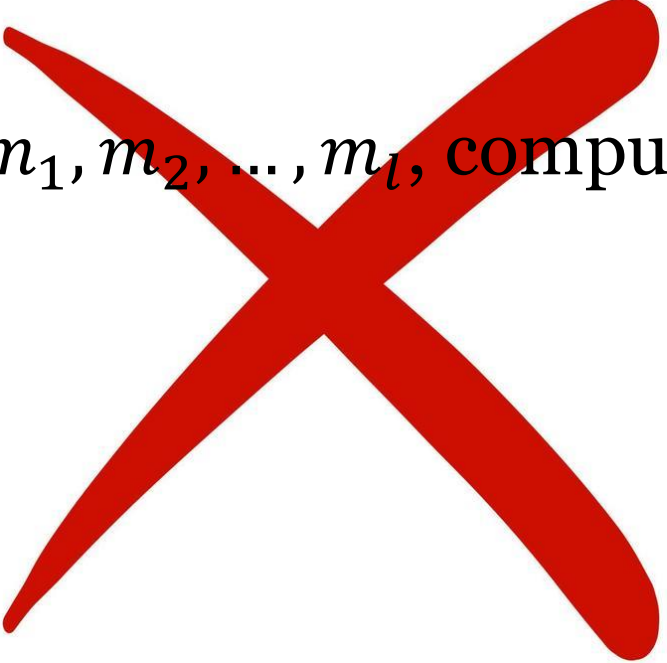
Longer messages?

Idea 2: parse m as $m_1, m_2, \dots, m_l$, compute $t_i = F_k(i || m_i)$



Longer messages?

Idea 3: parse m as $m_1, m_2, \dots, m_l$, compute $t_i = F_k(l||i||m_i)$



Longer messages?

Idea 4: parse m as $m_1, m_2, \dots, m_l$, compute $t_i = F_k(r || l || i || m_i)$

Send $(r, t_1, \dots, t_l)$ as the tag

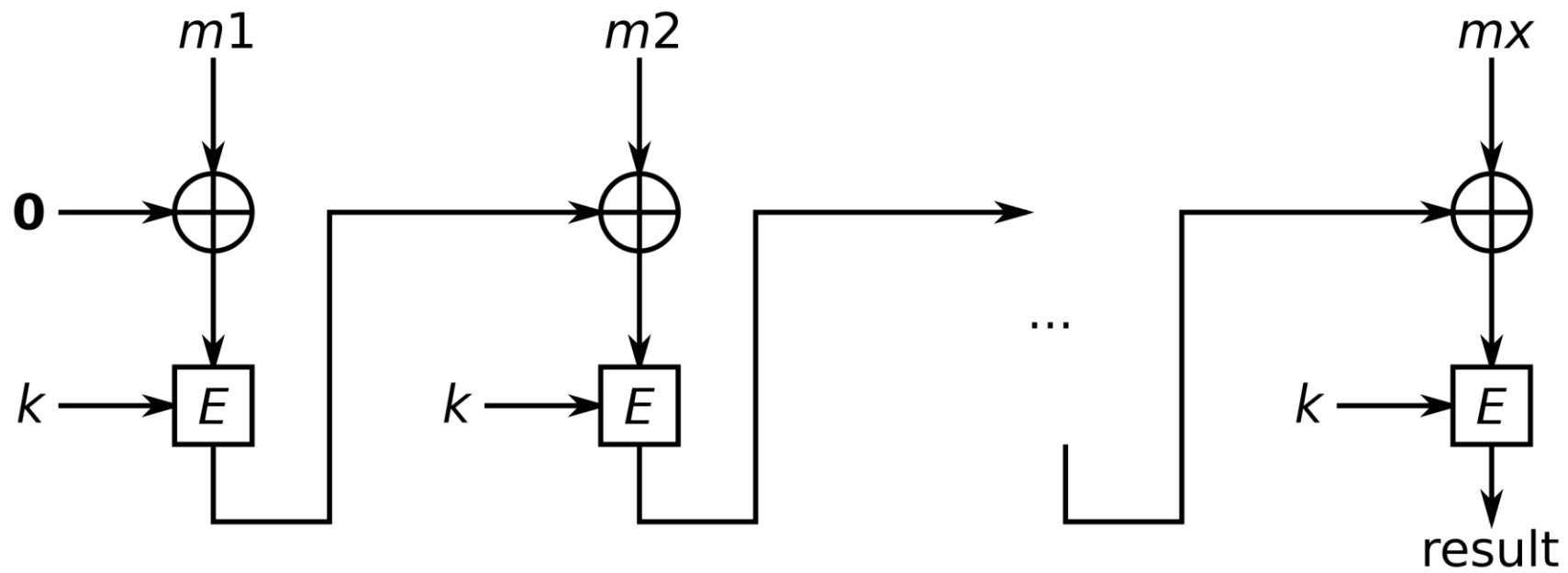
It is a secure tag. Proof.

Two events: Repeat r or a new block contained in Adv's message

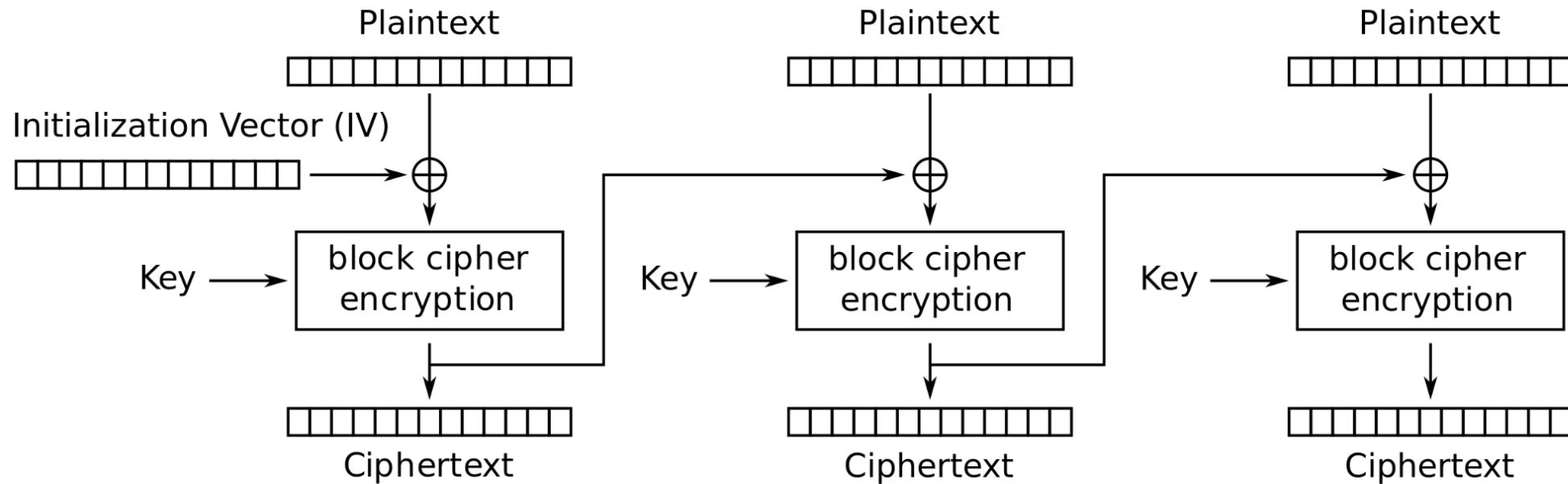
MAC in practice

- Can be deterministic
- Can be shorter than the message

CBC-MAC



CBC-mode encryption

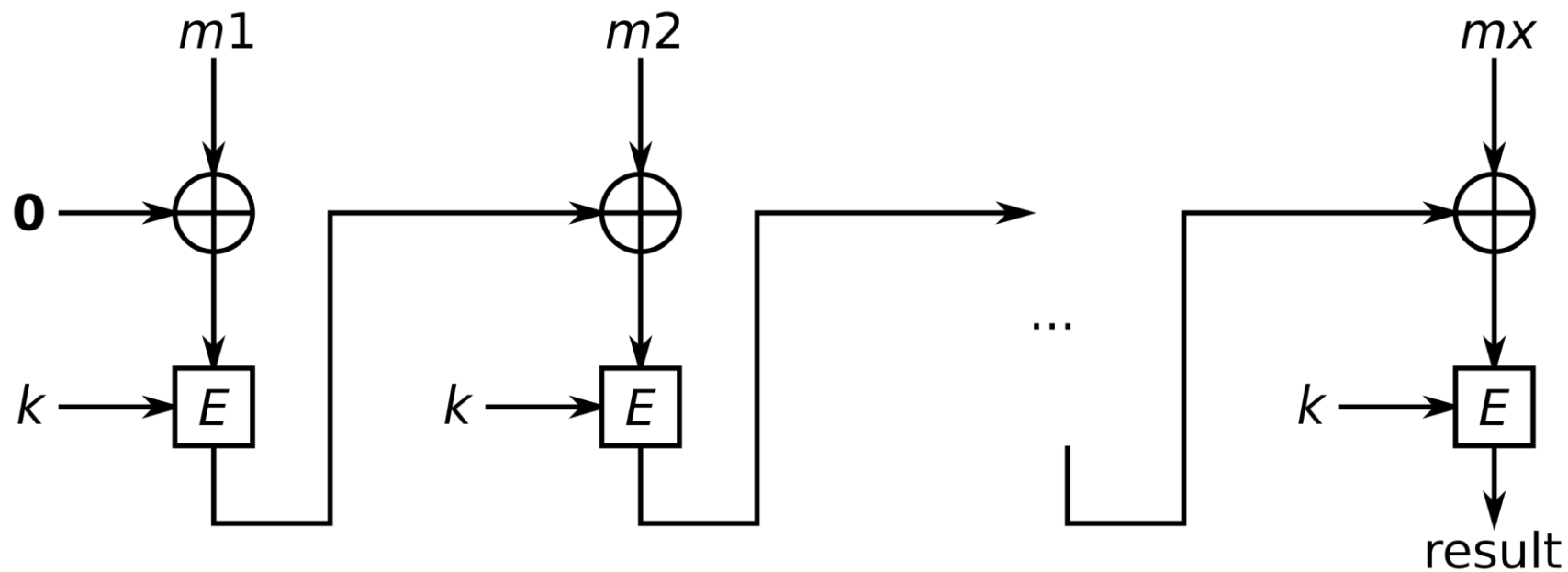


Cipher Block Chaining (CBC) mode encryption

CBC MAC vs. CBC encryption

1. CBC MAC is deterministic, IV has to be 0
 - Not secure if IV is randomized
2. t is shorter than the message
 - Not secure if we include the output of every block cipher encryption

CBC-MAC



Still only secure for fixed length message!!

Length extension attack

CBC-MAC for arbitrary-length messages

1. Prepend the message length $|m|$ as an n-bit string
 - What about append? (Bonus question)

